

Climate 5000 Series - Gen 4

24K BTU Single Zone Max Performance System

4-Way Cassette IDU

Engineering
Submittal
Sheet



BOSCH

Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

24K Btu Single Zone Max Performance System - 4-Way Cassette IDU

Indoor part number	8-733-962-681
Indoor model number	BMS500-AAU024-1AHCXD
Outdoor part number	8-733-962-691
Outdoor model number	BMS500-AAS024-1CSXHD

Efficiency Data AHRI 210/240 2023

SEER2	20.7
EER2	12.0
HSPF2 Region IV	11.6
HSPF2 Region V	9.2
COP(47F)	3.54
COP (5F)	2.23

Performance Data AHRI 210/240 2023

Cooling	Capacity Range	12800~26200
	Rated Capacity	23500
	Current	8.9
	Input Power	1958
Heating	Capacity Range	11500~30700
	Rated Capacity (47F)	25000
	Rated Capacity (5F)	23800
	Current	9.5
	Input Power	2070

Electrical Data

Power supply (V / Ø / Hz)	208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)	3.0 / 24.9
Max. fuse indoor/outdoor (A)	15.0 / 25.0

Compressor Data

Type:	Rotary
Refrigerant oil /oil charge (ml)	VG74 / 620
Input (W)	805 / 2045
Rated current (RLA/A)	4.15 / 9.30

Piping

Refrigerant precharge (oz/kg)	63.49 / 1.80
Length for precharge (ft/m)	24.6 / 7.5
Liquid side (in/mm)	3/8" / Ø9.52
Gas side (in/mm)	5/8" / Ø15.9
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.32 / 30
Max. refrigerant pipe length (ft/m)	164.00 / 50
Max. difference in height (ft/m)	82.00 / 25

Operation Range

ODU	Cooling (°F / °C)	-22~122 / -30~50
	Heating (°F / °C)	-22~75 / -30~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		4.98
IDU	Fan motor model	ZKFN-45-8-1
	Fan motor qty (ea)	1
	Fan motor input (W)	50.0
	Fan speed (RPM)	624 / 568 / 426
	Air flow, Turbo / H / M / L / Si (CFM)	765.2 / 706.3 / 635.7 / 547.4 / N/A
	Approx. Air Throw Range ft (m)	13.1 / 4
	Noise level, Turbo / H / M / L / Si (dB(A))	48.5 / 45.0 / 42.5 / 33.5 / N/A
	Net weight (lbs / kg)	47.18 / 21.4
	Gross weight (lbs / kg)	59.08 / 26.8
	Panel net weight (lbs / kg)	13.23 / 6
ODU	Gross panel weight (lbs / kg)	19.84 / 9
	Drain pipe diameter (mm)	ODΦ25mm
	Fan motor model	ZKFN-120-8-2
	Fan motor qty (ea)	1
	Fan motor input (W)	150
	Fan motor speed (RPM)	950 / 800 / 500
	Air flow, Max (CFM)	2236.7
	Noise level, Max (dB(A))	61.5
	Net weight (lbs / kg)	130.29 / 59.1
	Gross weight (lbs / kg)	141.31 / 64.1

Accessories

BMS-RCG-XXD (8733962703)	Decorative Grille - 24K ~ 48K
BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK3-7X1 (8733951015)	25ft Lineset

Standard Parts

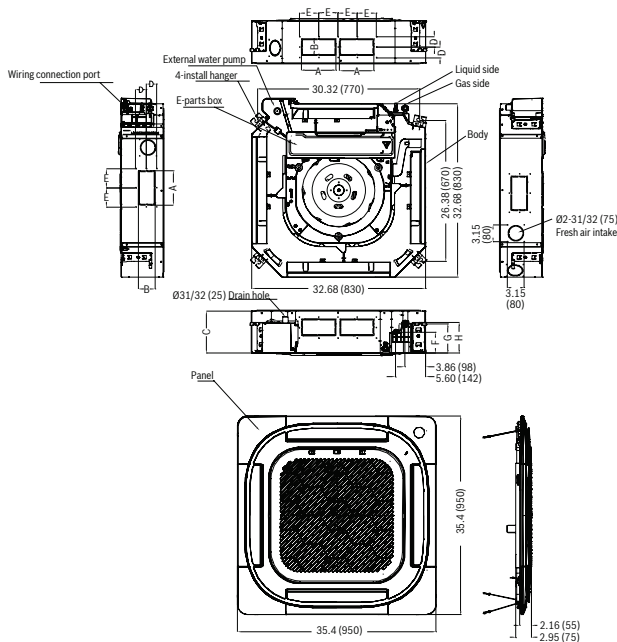
8-733-969-065	Remote Control
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Climate 5000 Series - Gen 4

24K Btu Single Zone Max Performance System - 4-Way Cassette IDU

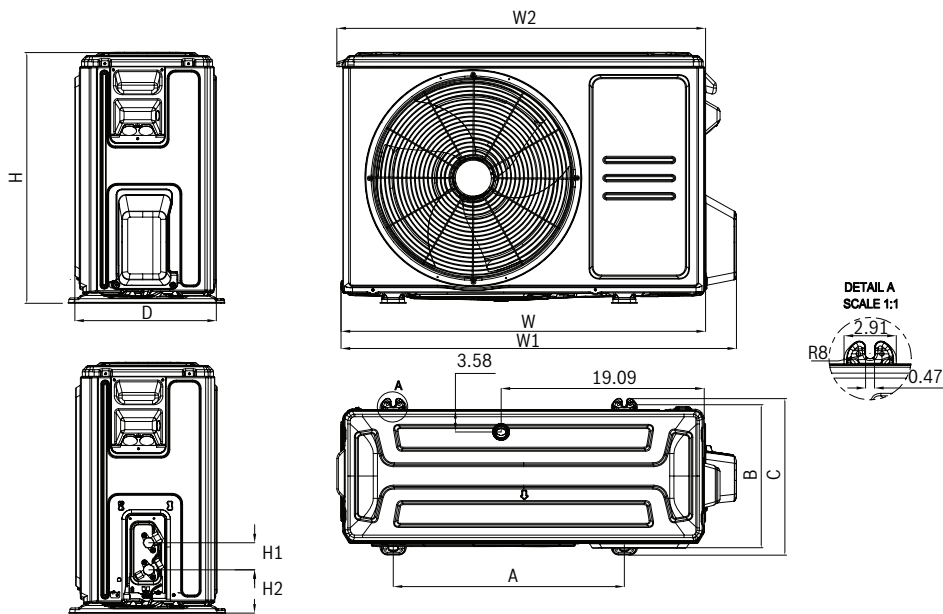


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU024-1AHCXD	32.68	32.68	8.07

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	A (in)	B (in)	C (in)	H1 (in)	H2 (in)
BMS500-AAS024-1CSXHD	37.24	16.14	31.89	40.55	37.24	26.5	15.87	17.9	2.36	4.25

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24K Btu Single Zone Max Performance System - 4-Way Cassette IDU


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Cooling Performance - Indoor Dry Bulb Temperature at 765 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	16478	16501	16518	16519	16541	17471	17492	17509	17513	17534	18515	18539	18560	18563	18584	19631	19652	19674	19677	19701
	S/T	0.60	0.71	0.80	0.80	0.91	0.57	0.67	0.76	0.76	0.86	0.53	0.63	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.77
	PI	1.17	1.17	1.17	1.17	1.18	1.24	1.24	1.24	1.24	1.25	1.32	1.32	1.32	1.32	1.32	1.39	1.40	1.40	1.40	1.40
-13°F (-25.0°C)	TC	17852	17876	17895	17896	17920	18927	18950	18969	18972	18995	20058	20084	20107	20110	20133	21267	21290	21313	21316	21343
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.68	0.68	0.78
	PI	1.21	1.22	1.22	1.22	1.22	1.29	1.29	1.29	1.29	1.29	1.36	1.37	1.37	1.37	1.37	1.45	1.45	1.45	1.45	1.45
-4°F (-20.0°C)	TC	19225	19251	19271	19272	19298	20383	20408	20428	20432	20456	21601	21629	21653	21657	21682	22903	22928	22953	22956	22985
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.79
	PI	1.26	1.26	1.26	1.26	1.26	1.33	1.34	1.34	1.34	1.34	1.41	1.41	1.42	1.42	1.42	1.50	1.50	1.50	1.50	1.50
0°F (-17.8°C)	TC	19835	19862	19883	19884	19911	21030	21055	21076	21080	21105	22287	22316	22341	22345	22370	23630	23655	23681	23685	23714
	S/T	0.63	0.73	0.84	0.84	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.65	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	1.28	1.28	1.28	1.28	1.28	1.35	1.36	1.36	1.36	1.36	1.43	1.44	1.44	1.44	1.44	1.52	1.52	1.52	1.52	1.53
5°F (-15.0°C)	TC	20598	20626	20648	20649	20677	21839	21865	21887	21891	21917	23144	23174	23200	23204	23230	24539	24565	24592	24596	24626
	S/T	0.63	0.74	0.84	0.84	0.96	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81
	PI	1.30	1.30	1.30	1.30	1.31	1.38	1.38	1.38	1.38	1.38	1.46	1.46	1.47	1.47	1.47	1.55	1.55	1.55	1.55	1.56
14°F (-10.0°C)	TC	24025	24058	24083	24085	24117	25473	25503	25528	25534	25564	26995	27030	27060	27065	27095	28621	28653	28684	28688	28724
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79
	PI	1.05	1.05	1.05	1.05	1.05	1.11	1.11	1.11	1.11	1.12	1.18	1.18	1.18	1.18	1.18	1.25	1.25	1.25	1.25	1.25
17°F (-8.3°C)	TC	25168	25201	25229	25230	25264	26684	26716	26742	26748	26779	28279	28315	28347	28352	28384	29982	30015	30048	30052	30090
	S/T	0.62	0.72	0.82	0.82	0.94	0.58	0.68	0.78	0.78	0.88	0.55	0.64	0.73	0.73	0.83	0.52	0.61	0.69	0.69	0.79
	PI	0.96	0.97	0.97	0.97	0.97	1.02	1.02	1.02	1.02	1.03	1.08	1.08	1.09	1.09	1.09	1.15	1.15	1.15	1.15	1.15
23°F (-5.0°C)	TC	24676	24709	24736	24737	24770	26163	26194	26220	26225	26256	27726	27762	27793	27798	27829	29397	29429	29461	29465	29502
	S/T	0.61	0.72	0.81	0.81	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.72	0.72	0.83	0.51	0.60	0.68	0.68	0.78
	PI	0.91	0.91	0.91	0.91	0.92	0.97	0.97	0.97	0.97	0.97	1.03	1.03	1.03	1.03	1.03	1.09	1.09	1.09	1.09	1.09
35°F (1.7°C)	TC	23692	23724	23750	23751	23783	25120	25150	25175	25180	25210	26621	26656	26685	26690	26720	28225	28256	28286	28291	28326
	S/T	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.76
	PI	0.81	0.81	0.81	0.81	0.81	0.86	0.86	0.86	0.86	0.86	0.91	0.91	0.91	0.91	0.91	0.96	0.97	0.97	0.97	0.97
41°F (5.0°C)	TC	24999	25033	25060	25061	25095	26506	26537	26563	26569	26600	28089	28126	28157	28162	28194	29782	29814	29847	29851	29888
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.69	0.69	0.78
	PI	1.07	1.07	1.07	1.07	1.07	1.13	1.13	1.13	1.13	1.13	1.20	1.20	1.20	1.20	1.20	1.27	1.27	1.27	1.27	1.27
47°F (8.3°C)	TC	26306	26341	26370	26371	26407	27891	27924	27952	27958	27991	29558	29596	29629	29635	29668	31339	31373	31407	31412	31451
	S/T	0.63	0.73	0.83	0.83	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.65	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	1.32	1.32	1.33	1.33	1.33	1.40	1.40	1.41	1.41	1.41	1.49	1.49	1.49	1.49	1.49	1.58	1.58	1.58	1.58	1.58
50°F (10.0°C)	TC	26268	26304	26332	26333	26369	27851	27884	27912	27917	27950	29515	29554	29587	29592	29625	31294	31328	31362	31367	31406
	S/T	0.62	0.73	0.82	0.82	0.94	0.58	0.68	0.78	0.78	0.89	0.55	0.65	0.73	0.73	0.84	0.52	0.61	0.69	0.69	0.79
	PI	1.36	1.36	1.37	1.37	1.37	1.44	1.45	1.45	1.45	1.45	1.53	1.53	1.53	1.53	1.54	1.62	1.62	1.63	1.63	1.63
59°F (15.0°C)	TC	26155	26190	26218	26219	26255	27731	27764	27791	27797	27830	29388	29426	29459	29464	29497	31159	31192	31226	31231	31270
	S/T	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.76
	PI	1.48	1.48	1.48	1.48	1.49	1.57	1.57	1.57	1.57	1.58	1.66	1.67	1.67	1.67	1.67	1.76	1.77	1.77	1.77	1.77
68°F (20.0°C)	TC	26041	26077	26105	26106	26141	27611	27644	27671	27676	27709	29260	29298	29331	29337	29369	31023	31057	31091	31096	31135
	S/T	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.72	0.72	0.83	0.51	0.60	0.68	0.68	0.78	0.48	0.57	0.65	0.65	0.74
	PI	1.60	1.60	1.60	1.60	1.61	1.70	1.70	1.70	1.70	1.70	1.80	1.80	1.80	1.80	1.80	1.90	1.91	1.91	1.91	1.91
77°F (25.0°C)	TC	25928	25963	25991	25992	26027	27491	27523	27550	27556	27589	29133	29171	29203	29209	29241	30888	30922	30956	30960	30999
	S/T	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.80	0.49	0.58	0.66	0.66	0.75	0.47	0.55	0.62	0.62	0.71
	PI	1.72	1.72	1.72	1.72	1.72	1.82	1.82	1.83	1.83	1.83	1.93	1.93	1.93	1.93	1.94	2.05	2.05	2.05	2.05	2.05
82°F (27.8°C)	TC	25548	25583	25610	25611	25646	27088	27120	27147	27152	27184	28706	28744	28776	28781	28813	30436	30469	30502	30507	30545
	S/T	0.56	0.66	0.75	0.75	0.85	0.53	0.62	0.70	0.70	0.80	0.50	0.58	0.66	0.66	0.76	0.47	0.55	0.63	0.63	0.71
	PI	1.79	1.80	1.80	1.80	1.80	1.90	1.91	1.91	1.91	1.91	2.02	2.02	2.02	2.02	2.02	2.14	2.14	2.14	2.14	2.15
86°F (30.0°C)	TC	24845	24878	24905	24906	24940	26342	26373	26399	26404	26436	27916	27952	27983	27988	28020	29548	29580	29612	29667	29704
	S/T	0.58	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79	0.49	0.58	0.65	0.65	0.75
	PI	1.87	1.87	1.88	1.88	1.88	1.98	1.99	1.99	1.99	1.99	2.10	2.10	2.11	2.11	2.11	2.23	2.23	2.23	2.23	2.24
95°F (35.0°C)	TC	23262	23293	23318	23319	23350	24663</														

Climate 5000 Series - Gen 4

24K Btu Single Zone Max Performance System - 4-Way Cassette IDU



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Heating Performance - Indoor Dry Bulb Temperature at 765 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	9743	9548	9500	9462	9377	2.38	2.40	2.40	2.41	2.41
-13°F(-25.0°C)	12922	12663	12600	12550	12437	2.45	2.47	2.48	2.48	2.49
-4°F(-20.0°C)	17742	17387	17300	17231	17076	2.71	2.74	2.74	2.75	2.76
0°F(-17.8°C)	20704	20290	20189	20108	19927	2.88	2.91	2.91	2.92	2.93
5°F(-15.0°C)	24408	23920	23800	23705	23491	3.10	3.12	3.13	3.13	3.15
14°F(-10.0°C)	24485	23995	23875	23780	23565	2.87	2.89	2.90	2.91	2.92
17°F(-8.3°C)	24510	24020	23900	23804	23590	2.80	2.82	2.82	2.83	2.84
22°F(-5.6°C)	25380	24873	24748	24649	24428	2.84	2.86	2.87	2.87	2.89
27°F(-2.8°C)	26250	25725	25597	25494	25265	2.88	2.91	2.91	2.92	2.93
32°F(0.0°C)	27120	26578	26445	26339	26102	2.93	2.95	2.95	2.96	2.97
35°F(1.7°C)	27642	27090	26954	26846	26605	2.95	2.98	2.98	2.99	3.00
37°F(2.8°C)	28283	27717	27578	27468	27221	2.87	2.89	2.90	2.91	2.92
42°F(5.6°C)	29883	29286	29139	29023	28761	2.67	2.69	2.69	2.70	2.71
44.6°F(7.0°C)	30524	29913	29764	29644	29378	2.59	2.61	2.61	2.62	2.63
47°F(8.3°C)	31484	30854	30700	30577	30302	2.47	2.49	2.49	2.49	2.51
52°F(11.1°C)	31110	30488	30336	30214	29942	2.22	2.24	2.24	2.25	2.26
57°F(13.9°C)	30737	30122	29971	29851	29583	1.97	1.99	1.99	2.00	2.01
59°F(15.0°C)	30587	29975	29825	29706	29439	1.88	1.89	1.89	1.90	1.91
62°F(16.7°C)	30363	29756	29607	29488	29223	1.73	1.74	1.75	1.75	1.76
64.4°F(18.0°C)	30183	29580	29432	29314	29050	1.61	1.62	1.63	1.63	1.64
69.8°F(21.0°C)	29780	29184	29038	28922	28662	1.34	1.36	1.36	1.36	1.37
75.2°F(24.0°C)	29376	28789	28645	28530	28273	1.08	1.09	1.09	1.09	1.10

Bolded values were tested. All other values are simulated and for reference only.